



Bhabha University, Bhopal (M.P.)

Scheme of Examination

Program: Bachelor of Technology Branch: Computer Science and Engineering Session: 2020-21

Semester : VII Specialization: Computer Science and Engineering

S.No .	Subject Code	Subject Name	Maximum Marks Allotted					Total Marks	Contact Hours per week			Total Credits
			Theory			Practical			L	T	P	
			End Sem.	Mid Sem. Exam.	Quiz/ Assignment	End Sem	Termwork					
							Lab Work & Sessional					
1.	CS-71	Software Architectures	70	20	10	30	20	150	3	1	2	5
2.	CS-72	Compiler Design	70	20	10	30	20	150	3	1	2	5
3.	CS-73	Elective-I	70	20	10	-	-	100	4	-	-	4
4.	CS-74	Elective-II	70	20	10			100	4	-	-	4
5.	CS-75	Major Project I	-	-	20	60	20	100	-	-	6	3
6.	CS-76	Industrial-Training	-	-	20	30	50	100	-	-	4	2
7	CS-77	Creativity & Entrepreneurship Development (Internal assessment)					50	50			2	1
		Total	280	80	80	150	160	750	14	2	16	24

***End Semester Examination will be of three hours**

Elective I	Elective II
CS -73 (A) Computational Intelligence	CS-74(A) Cryptography & Information Security
CS-73(B) Deep & Reinforcement Learning	CS-74(B) Digital Image Processing
CS-73 (C) Wireless & Mobile Computing	CS74(C) Data mining and Warehousing

PRACTICALS SUBJECT CS- 71: Software Architectures
 CS-75: Major Project-I
 CS-77: Creativity & Entrepreneurship Development

CS-72 : Compiler Design
 CS-76 : Industrial Training

1 Hr Lecture	1 Hr Tutorial	2 Hr Practical
1 Credit	1 Credit	1 Credit



BHABHA UNIVERSITY, BHOPAL

**COURSE CURRICULUM
FACULTY OF ENGINEERING**

Undergraduate Programme

(Courses effective from Academic Year 2020-21)

**SYLLABUS OF COURSES TO BE OFFERED
“Bachelor of Technology”**

Disclaimer: The undergraduate programme in Bachelor of Technology syllabus & scheme is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

Dean faculty of Engineering, Bhabha University

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CS-71 Software Architectures

Unit 1. Overview of Software development methodology and software quality model, different models of software development and their issues. Introduction to software architecture, evolution of software architecture, software components and connectors, common software architecture frameworks, Architecture business cycle – architectural patterns – reference model.

Unit 2. Software architecture models: structural models, framework models, dynamic models, process models. Architectures styles: dataflow architecture, pipes and filters architecture, call-and return architecture, data-centered architecture, layered architecture, agent based architecture, Micro-services architecture, Reactive Architecture, Representational state transfer architecture etc.

Unit 3. Software architecture implementation technologies: Software Architecture Description Languages (ADLs), Struts, Hibernate, Node JS, Angular JS, J2EE – JSP, Servlets, EJBs; middleware: JDBC, JNDI, JMS, RMI and CORBA etc. Role of UML in software architecture.

Unit4. Software Architecture analysis and design: requirements for architecture and the life-cycle view of architecture design and analysis methods, architecture-based economic analysis: Cost Benefit Analysis Method (CBAM), Architecture Tradeoff Analysis Method (ATAM). Active Reviews for Intermediate Design (ARID), Attribute Driven Design method (ADD), architecture reuse, Domain –specific Software architecture.

Unit 5. Software Architecture documentation: principles of sound documentation, refinement, context diagrams, variability, software interfaces. Documenting the behavior of software elements and software systems, documentation package using a seven-part template.

LIST OF EXPERIMENTS

- 1) To Describe structural model and its framework.
- 2) To Write a architecture style & its data flow architecture.
- 3) To Describe pipe & filter architecture.
- 4) To Write a cost benefit analysis method (CBAM)
- 5) To Write a Architecture trade off analysis methide (ATAM).
- 6) To Describe evaluation of software architecture
- 7) To Describe J2EE, JSP & Servlets



REFERENCES:

1. Bass, L., P. Clements, and R. Kazman, “Software Architecture in Practice”, Second Edition, Prentice- Hall.
2. Jim Keogh, “J2EE – Complete Reference”, Tata McGrawHill.
3. Dikel, David, D. Kane, and J. Wilson, “Software Architecture: Organizational Principles and Practices”, Prentice-Hall.
4. Bennett, Douglas, “Designing Hard Software: The Essential Tasks”, Prentice-Hall, 1997.
5. Clements, Paul, R. Kazman, M. Klein, “Evaluating Software Architectures: Methods and Case Studies”, Addison Wesley, 2001.
6. Albin, S. “The Art of Software Architecture”, Indiana: Wiley, 2003.
7. Robert Mee, and Randy Stafford, “Patterns of Enterprise Application Architecture”, Addison-Wesley, 2002.
8. Witt, B., T. Baker and E. Meritt, “Software Architecture and Design: Principles, Models and Methods”



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CS-72 Compiler Design

Unit-I Introduction to compiling & Lexical Analysis

Introduction of Compiler, Major data Structure in compiler, BOOT Strapping & Porting, Compiler structure: analysis-synthesis model of compilation, various phases of a compiler, Lexical analysis: Input buffering , Specification & Recognition of Tokens, LEX.

Unit-II Syntax Analysis & Syntax Directed Translation

Syntax analysis: CFGs, Topdown parsing, Brute force approach, recursive descent parsing, transformation on the grammars, predictive parsing, bottom up parsing, operator precedence parsing, LR parsers (SLR, LALR, LR), Parser generation. Syntax directed definitions: Construction of Syntax trees, Bottom up evaluation of S-attributed definition, L-attributed definition, Topdown translation, Bottom Up evaluation of inherited attributes Recursive Evaluation, Analysis of Syntax directed definition.

Unit-III Type Checking & Run Time Environment

Type checking: type system, specification of simple type checker, equivalence of expression, types, type conversion, overloading of functions and operations, polymorphic functions. Run time Environment: storage organization, Storage allocation strategies, parameter passing, dynamic storage allocation, Symbol table

Unit –IV Code Generation

Intermediate code generation: Declarations, Assignment statements, Boolean expressions, Case statements, Back patching, Procedure calls Code Generation: Issues in the design of code generator, Basic block and flow graphs, Register allocation and assignment, DAG representation of basic blocks, peephole optimization, generating code from DAG.

Unit –V Code Optimization

Introduction to Code optimization: sources of optimization of basic blocks, loops in flow graphs, dead code elimination, loop optimization, Introduction to global data flow analysis, Code Improving transformations, Data flow analysis of structure flow graph Symbolic debugging of optimized code.



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LIST OF EXPERIMENTS:

- ☐ Develop a lexical analyzer to recognize a few patterns.
- ☐ Write a programme to parse using Brute force technique of Topdown parsing.
- ☐ Develop LL (1) parser (Construct parse table also).
- ☐ Develop an operator precedence parser (Construct parse table also)
- ☐ Develop a recursive descent parser
- ☐ Write a program for generating for various intermediate code forms
 - a. Three address code ii) Polish notation
- ☐ Write a program to simulate Heap storage allocation strategy
- ☐ Generate Lexical analyzer using LEX
- ☐ Generate YACC specification for a few syntactic categories.
- ☐ Given any intermediate code form implement code optimization techniques
- ☐ Study of an Object Oriented Compiler.

REFERENCES

1. A. V. Aho, R. Sethi, and J. D. Ullman. Compilers: Principles, Techniques and Tools, Pearson Education
2. Raghavan, Compiler Design, TMH Pub.
3. Loudon. Compiler Construction: Principles and Practice, Cengage Learning
4. A. C. Holub. Compiler Design in C, Prentice-Hall Inc., 1993.
5. Mak, writing compiler & Interpreters, Wiley Pu





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CS -73 (A) Computational Intelligence

Unit1: Introduction to Computational Intelligence; types of Computational Intelligence, components of Computational Intelligence. Concept of Learning/Training model. Parametric Models, Nonparametric Models. Multilayer Networks: Feed Forward network, Feedback network.

Unit2. Fuzzy Systems: Fuzzy set theory: Fuzzy sets and operations, Membership Functions, Concept of Fuzzy relations and their composition, Concept of Fuzzy Measures; Fuzzy Logic: Fuzzy Rules, Inferencing; Fuzzy Control - Selection of Membership Functions, Fuzzyfication, Rule Based Design & Inferencing, Defuzzification.

Unit3. Genetic Algorithms: Basic Genetics, Concepts, Working Principle, Creation of Offsprings, Encoding, Fitness Function, Selection Functions, Genetic Operators-Reproduction, Crossover, Mutation; Genetic Modeling, Benefits.

Unit4. Rough Set Theory - Introduction, Fundamental Concepts, Set approximation, Rough membership, Attributes, Optimization. Hidden Markov Models, Decision tree model.

Unit5. Introduction to Swarm Intelligence, Swarm Intelligence Techniques: Ant Colony Optimization, Particle Swarm Optimization, Bee Colony Optimization etc. Applications of Computational Intelligence.

REFERENCES

1. Russell C. Eberhart and Yuhui Shi, Computational Intelligence: Concepts to Implementations, Morgan Kaufmann Publishers.
2. Andries P. Engelbrecht, Computational Intelligence: An Introduction, Wiley Publishing.
3. Simon Haykin, Neural Networks: A Comprehensive Foundation, Prentice Hall.
4. David E. Goldberg, Genetic Algorithm in Search Optimization and Machine Learning, Pearson Education.
5. Jagdish Chand Bansal, Pramod Kumar Singh, Nikhil R. Pal, Evolutionary and Swarm Intelligence Algorithms, Springer Publishing, 2019.
6. S. Rajeskar, G.A. Vijayalakshmi Pai, "Neural Networks, Fuzzy Logic, Genetic Algorithms Synthesis and Applications".
7. J.S. Roger Jang, C.T. Sun, E. Mizutani, "Neuro-Fuzzy and Soft Computing: A Computational Approach to Learning & Machine Intelligence", PHI, 2002.



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CS-73(B) Deep & Reinforcement Learning

Unit 1: History of Deep Learning, McCulloch Pitts Neuron, Thresholding Logic, Activation functions, Gradient Descent (GD), Momentum Based GD, Nesterov Accelerated GD, Stochastic GD, AdaGrad, RMSProp, Adam, Eigenvalue Decomposition. Recurrent Neural Networks, Backpropagation through time (BPTT), Vanishing and Exploding Gradients, Truncated BPTT, GRU, LSTMs, Encoder Decoder Models, Attention Mechanism, Attention overimages.

Unit2: Autoencoders and relation to PCA, Regularization in autoencoders, Denoising autoencoders, Sparse autoencoders, Contractive autoencoders, Regularization: Bias Variance Tradeoff, L2 regularization, Early stopping, Dataset augmentation, Parameter sharing and tying, Injecting noise at input, Ensemble methods, Dropout, Batch Normalization, Instance Normalization, Group Normalization.

Unit 3: Greedy Layerwise Pre-training, Better activation functions, Better weight initialization methods, Learning Vectorial Representations Of Words, Convolutional Neural Networks, LeNet, AlexNet, ZF-Net, VGGNet, GoogLeNet, ResNet, Visualizing Convolutional Neural Networks, Guided Backpropagation, Deep Dream, Deep Art, Recent Trends in Deep Learning Architectures.

Unit 4: Introduction to reinforcement learning (RL), Bandit algorithms – UCB, PAC, Median Elimination, Policy Gradient, Full RL & MDPs, Bellman Optimality, Dynamic Programming - Value iteration, Policy iteration, and Q-learning & Temporal Difference Methods, Temporal-Difference Learning, Eligibility Traces, Function Approximation, Least Squares Methods

Unit 5: Fitted Q, Deep Q-Learning, Advanced Q-learning algorithms, Learning policies by imitating optimal controllers, DQN & Policy Gradient, Policy Gradient Algorithms for Full RL, Hierarchical RL, POMDPs, Actor-Critic Method, Inverse reinforcement learning, Maximum Entropy Deep Inverse Reinforcement Learning, Generative Adversarial Imitation Learning, Recent Trends in RL Architectures.

REFERENCES

1. Deep Learning, An MIT Press book, Ian Goodfellow and Yoshua Bengio and Aaron Courville
2. Pattern Classification- Richard O. Duda, Peter E. Hart, David G. Stork, John Wiley & Sons Inc.
3. Reinforcement Learning: An Introduction, Sutton and Barto, 2nd Edition.
4. Reinforcement Learning: State-of-the-Art, Marco Wiering and Martijn van Otterlo, Eds



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CS-73 (C) Wireless & Mobile Computing

Unit 1: Review of traditional networks: Review of LAN, MAN, WAN, Intranet, Internet, and interconnectivity devices: bridges, Routers etc. Review of TCP/IP Protocol Architecture: ARP/RARP, IP addressing, IP Datagram format and its Delivery, Routing table format, ICMP Messages, Subnetting, Supernetting and CIDR, DNS. NAT: Private addressing and NAT, SNAT, DNAT, NAT and firewalls, VLANs: Concepts, Comparison with Real LANS, Type of VLAN, Tagging, IPV6: address structure, address space and header.

Unit2: Study of traditional routing and transport: Routing Protocols: BGP- Concept of hidden network and autonomous system, An Exterior gateway protocol, Different messages of BGP. Interior Gateway protocol: RIP, OSPF. Multiplexing and ports, TCP: Segment format, Sockets, Synchronization, Three Way Hand Shaking, Variable window size and Flow control, Timeout and Retransmission algorithms, Connection Control, Silly window Syndrome. Example of TCP: Tahoe, Reno, Sack etc. UDP: Message Encapsulation, Format and Pseudoheader.

Unit 3: Wireless LAN: Transmission Medium For WLANs, MAC problems, Hidden and Exposed terminals, Near and Far terminals, Infrastructure and Ad hoc Networks, IEEE 802.11- System arch, Protocol arch, Physical layer, Concept of spread spectrum, MAC and its management, Power management, Security. Mobile IP: unsuitability of Traditional IP; Goals, Terminology, Agent advertisement and discovery, Registration, Tunneling techniques. Ad hoc network routing: Ad hoc Network routing v/s Traditional IP routing, types of routing protocols, Examples: OADV, DSDV, DSR, ZRP etc.

Unit 4: Mobile transport layer: unsuitability of Traditional TCP; I-TCP, S-TCP, M-TCP. Wireless Cellular networks: Cellular system, Cellular networks v/s WLAN, GSM – Services, system architecture, Localization and calling, handover and Roaming.

Unit 5: Mobile Device Operating Systems: Special Constraints & Requirements, Commercial Mobile Operating Systems. Software Development Kit: iOS, Android etc. MCommerce : Structure , Pros & Cons, Mobile Payment System , Security Issues

REFERENCE

1. W. Richard Stevens, "TCP/IP Illustrated Vol-I", Addison-Wesley.
2. C.K. Toh, "AdHoc Mobile Wireless Networks", First Edition, Pearson Education.
3. Uwe Hansmann, Lothar Merk, Martin S. Nicklons and Thomas Stober, "Principles of Mobile Computing", Springer
4. Android Developers : <http://developer.android.com/index.html>
5. Apple Developer : <https://developer.apple.com/>
6. Windows Phone Dev Center : <http://developer.windowsphone.com/>
7. BlackBerry Developer : <http://developer.blackberry.com/>.



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CS-74(A) Cryptography & Information Security

UNIT I: Mathematical Background for Cryptography: Abstract Algebra, Number Theory, Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler Phi-Function, Euler's theorem.

Introduction to Cryptography: Principles of Cryptography, Classical Cryptosystem, Cryptanalysis on Substitution Cipher (Frequency Analysis), Play Fair Cipher, Block Cipher. Data Encryption Standard (DES), Triple DES, Modes of Operation, Stream Cipher.

UNIT II: Advanced Encryption Standard (AES), Introduction to Public Key Cryptosystem, Discrete Logarithmic Problem, Diffie-Hellman Key Exchange Computational & Decisional Diffie-Hellman Problem, RSA Assumptions & Cryptosystem, RSA Signatures & Schnorr Identification Schemes, Primarily Testing, Elliptic Curve over the Reals, Elliptic curve Modulo a Prime., Chinese Remainder Theorem.

UNIT III: Message Authentication, Digital Signature, Key Management, Key Exchange, Hash Function. Universal Hashing, Cryptographic Hash Function, MD, Secure Hash Algorithm (SHA), Digital Signature Standard (DSS), Cryptanalysis: Time-Memory Trade-off Attack, Differential Cryptanalysis. Secure channel and authentication system like Kerberos.

UNIT IV: Information Security: Threats in Networks, Network Security Controls–Architecture, Wireless Security, Honey pots, Traffic Flow Security, Firewalls – Design and Types of Firewalls, Personal Firewalls, IDS, **Email Security:** Services Security for Email Attacks Through Emails, Privacy-Authentication of Source Message, Pretty Good Privacy (PGP), S-MIME. **IP Security:** Overview of IPsec, IP & IP version 6 Authentication, Encapsulation Security Payload ESP, Internet Key Exchange IKE, **Web Security:** SSL/TLS, Basic protocols of security. Encoding – Secure Electronic Transaction SET.

UNIT V: **Cryptography and Information Security Tools:** Spoofing tools: like Arping etc., **Footprinting Tools** (ex-nslookup, dig, Whois, etc.), **Vulnerabilities Scanning Tools** (i.e. Angry IP, HPing2, IP Scanner, Global Network Inventory Scanner, Net Tools Suite Pack.), NetBIOS Enumeration Using NetView Tool, **Steganography** Merge Streams, Image Hide, Stealth Files, Blindsiding: **STools, Steghide, Steganos.** Stegdetect, Steganalysis - Stego Watch- Stego Detection Tool, **StegSpy.** **Trojans Detection Tools** (i.e. Netstat, fPort, TCPView, CurrPorts Tool, Process Viewer), Lan Scanner Tools (i.e. look@LAN, Wireshark, Tcpdump). **DoS Attack Understanding Tools-** Jolt2, Bubonic.c, Land and LaTierra, Targa, Nemesis Blast, Panther2, Crazy Pinger, Some Trouble, UDP Flood, FSMax.

REFERENCES

1. Cryptography and Network Security Principles and Practice Fourth Edition, William Stallings, Pearson Education.
2. Network Security Essentials: Applications and Standards, by William Stallings. Prentice Hall.
3. Behrouz A Ferouzan, "Cryptography and Network Security" Tata Mc Graw Hills, 2007
4. Charles P. Fleegeer, Shari Lawrence Fleegeer "Security in Computing", 4th Edition Prentice Hall of India, 2006.
5. Introduction to Modern Cryptography by Jonathan Katz and Yehuda Lindell, Chapman and Hall/CRC.



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CS-74(B) Digital Image Processing

Unit-I: Digital Image fundamentals, A simple image model, Sampling and Quantization. Relationship between pixels. Imaging geometry. Image acquisition systems, Different types of digital images

Unit-II: Image transformations, Introduction to Fourier transforms, Discrete Fourier transforms, Fast Fourier transform, Walsh transformation, Hadamard transformation, Discrete Cosine Transformation.

Unit-III: Image enhancement, Filters in spatial and frequency domains, Histogram based processing. Image subtraction, Averaging, Image smoothing, Median filtering, Low pass filtering, Image sharpening by High pass filtering.

Unit-IV: Image encoding and segmentation, Encoding: Mapping, Quantizer, Coder. Error free compression, Lossy Compression schemes. JPEG Compression standard. Detection of discontinuation by point detection, Line detection, edge detection, Edge linking and boundary detection, Local analysis, Global processing via Hough transforms and graph theoretic techniques

Unit-V: Mathematical morphology- Binary, Dilation, crosses, Opening and closing, Simple methods of representation, Signatures, Boundary segments, Skeleton of a region, Polynomial approximation

REFERENCES

1. Rafael C Gonzalez, Richard E Woods 3rd Edition, Digital Image Processing Pearson.
2. Rafael C Gonzalez, Richard E Woods 3rd Edition, Digital Image Processing using Matlab– TMH.
3. Sonka, Digital Image Processing & Computer Vision , Cengage Learning
4. Jayaraman, Digital Image Processing, TMH.
5. Pratt, Digital Image Processing, Wiley India
6. Annadurai, Fundamentals of Digital Image Processing , Pearson Education .



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CS74(C) Data Mining & Warehousing

Unit-I: Data Warehousing: Introduction, Delivery Process, Data warehouse Architecture, Data Preprocessing: Data cleaning, Data Integration and transformation, Data reduction. Data warehouse Design: Datawarehouse schema, Partitioning strategy Data warehouse Implementation, Data Marts, Meta Data, Example of a Multidimensional Data model. Introduction to PatternWarehousing.

Unit-II: OLAP Systems: Basic concepts, OLAP queries, Types of OLAP servers, OLAP operations etc. Data Warehouse Hardware and Operational Design: Security, Backup And Recovery,

Unit-III: Introduction to Data & Data Mining : Data Types, Quality of data, Data Preprocessing, Similarity measures, Summary statistics, Data distributions, Basic data mining tasks, Data Mining V/s knowledge discovery in databases. Issues in Data mining. Introduction to Fuzzy sets and fuzzy logic.

Unit-IV: Supervised Learning: Classification: Statistical-based algorithms, Distance-based algorithms, Decision tree-based algorithms, Neural network-based algorithms, Rule-based algorithms, Probabilistic Classifiers

Unit-V: Clustering & Association Rule mining : Hierarchical algorithms, Partitional algorithms, Clustering large databases – BIRCH, DBSCAN, CURE algorithms. Association rules : Parallel and distributed algorithms such as Apriori and FP growth algorithms.

REFERENCES

1. Pang – ning Tan , Steinbach & Kumar, “*Introduction to Data Mining*”, Pearson Edu, 2019.
2. Jaiwei Han, Micheline Kamber, “*Data Mining : Concepts and Techniques*”, Morgan Kaufmann Publishers
3. Margaret H. Dunham, “*Data Mining : Introductory and Advanced topics*”, Pearson Edu., 2009.
4. Anahory & Murray, “*Data Warehousing in the Real World*”, Pearson Edu., 2009.



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CS-75 Major Project-I

Role of Major Projects in life of Engineering students are very crucial. It does not just affect the grades of Engineering but also matter a lot for good CV/Resume. So before choosing the major project, you should explore the options and pick the correct domain where the opportunities are immense.

Benefits of doing the Projects for Engineering students

Projects are generally done as a combined team effort. Two or more students work under a guide or a staff to get a certain results. By doing a project, you will

1. Understand your subject better
2. Get practical experience
3. Chance to showcase your skills
4. Learn about team work, communication skills and responsibilities

How to Choose the Best Projects

You can do any kind of projects based on your interests or subjects. The best way to go about this is to figure out what you are interested in. For example, if you are interested in Designing, then you can design a website of your own or create a poster using Photoshop or Illustrator. So, the first step is to find your interest and then do projects in your area of interest.

Major Highlight of the Program

The major highlight of the program are as following:

- Training in special skills on which the project will be developed
- IIT Mentors to help the individuals formajor projects of engineering
- Development of the project
- Boost in CV/Resume
- Job opportunities after engineering
- Industrial Connect



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CS-76 Industrial Training

Objective of Industrial Training

The objective of undertaking industrial training is to provide work experience so that student's engineering knowledge is enhanced and employment prospects are improved. The student should take this course as a window to the real World and should try to learn as much as possible from real life experiences by involving and interacting with industry staff. Industrial training also provides an opportunity to students to select an engineering problem and possibly an industry guide for their Major Project in final semester.

Scheme of Studies:

Duration: Minimum 2 weeks in summer break after VI semester, assessment to be done in VII Semester. Marks of various components in industry should be awarded by the I/c of training in Industry but in special circumstances if not awarded by the industry then faculty in charge /T.P.O. will give the marks. During training students will prepare a first draft of training report in consultation with section in charge. After training they will prepare final draft with the help of T.P.O. /Faculty of the Institute. Then they will present a seminar on their training and they will face viva-voce on training in the Institute.

Learning through Industrial Training

During industrial training students must observe following to enrich their learning:

- ☐ Industrial environment and work culture.
- ☐ Organizational structure and inter personal communication.
- ☐ Machines/equipment/instrument-their working and specifications.
- ☐ Product development procedure and phases.
- ☐ Project Planning, monitoring and control.
- ☐ Quality control and assurance.
- ☐ Maintenance system
- ☐ Costing system
- ☐ Stores and purchase systems.
- ☐ Layout of Computer/EDP/MIS centers.

Supervision of Industrial Training

Faculty and TPO are supposed to plan industrial training in such a manner that students get exposure on most of the above area in the field. One faculty member or TPO will plan industrial training of students in consultation with training manager of the industry (work place) as per the predefined objectives of training. Monitoring visits will be made by training and placement officer/faculty in-charge for the group of students, of the college during training.



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Guidance to the faculty / TPO for Planning and implementing the Industrial Training

Keeping in view the need of the contents, the industrial training program, which is spread to minimum 2 weeks duration, has to be designed in consultation with the authorities of the work place; Following are some of the salient points:

- ☐ Spelling out the objectives of the industrial training in behavioral terms and same is informed in advance to the 1) students, 2) authorities of the work place and 3) supervising faculty members.
- ☐ Discussing and preparing students for the training for which meetings with the students has to be planned.
- ☐ Meeting with industrial personnel and orienting them regarding the objective of the training and the expectations of the program.
- ☐ Correspondence with the authorities of the work place.
- ☐ Orientation classes for students on how to make the training most beneficial- monitoring daily diary, writing weekly reports, how to interact with various categories of industrial personnel, how to behave and undertake responsibilities, how to gather information from the workplace, ethics etc.
- ☐ Guiding students to make individual plans (week wise/ day wise) to undertake industrial training.
- ☐ Developing a system of maintaining training records, by teachers for every batch of students for convenient retrieval

Daily Diary- Industrial Training

Name of the Trainee----- College -----
Industry / work place ----- Week No-----
Department /Section ----- Date -----

Dates Brief of observations made, work done, problem/project undertaken, discussion held, literature consulted etc.

Signature of Supervisor Signature of Trainee Signature of Official in (TPO/Faculty) charge



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Action plan for planning stages at the Institutional Level

- Meeting with Principal
- Meeting with colleagues
- Correspondence with work place (Industry concerned)
- Meeting with authorities of workplace
- Orientation of students for industry training
- Scrutinizing individual training plan of students.
- Commencement of individual training
- First monitoring of industrial training
- Second monitoring of industrial training
- Finalization of Training report
- Evaluation of performance at industry level
- Evaluation of Industry Program in the Institutions.





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CS-77 Creativity & Entrepreneurship Development

Here, Students have to show their Creativity i.e. students have to develop new ideas and to discover new ways of looking at problems and opportunities. Students have to apply creative solutions to those problems and opportunities in order to enhance people's lives or to enrich society; referred as Innovation.

In Entrepreneurship Development, students have to apply those creative & innovative ideas to do something new in an efficient manner which will help them in campus selection, boost in CV/Resume and many more. Entrepreneurship is the result of a disciplined, systematic process of applying creativity and innovation to needs and opportunities in the marketplace.

Entrepreneurs are those who carry their creative ideas with the purposeful action and structure of a business. Researchers believe that entrepreneurs succeed by thinking and doing new things or old things in new ways.

